ARAB REPUBLIC OF EGYPT
MINISTRY OF EDUCATION & TECHNICAL EDUCATION
CENTRAL ADMINISTRATION FOR CURRICULUM DEVELOPMENT (CCIMD)

OFFICIAL SEAL
MOE APPROVED

OFFICIAL MINISTRY CONCEPTS & FORMULA COMPENDIUM

Authorized Ministry of Education Reference for General Secondary Exams

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 & EG-BACCALAUREATE EXAMINATIONS

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized official concepts compendium distributed to students inside ministerial examination halls, containing verified identities, integration forms, moment equilibrium rules, and physical constants.

CORE CURRICULUM HIGHLIGHTS:

- Exact ministerial formula tables used in official examination halls
- Trigonometric and inverse trigonometric standard derivatives
- Integration forms, substitution catalogs, and reduction formulas
- Friction, couples, pulley equations, and momentum conservation laws

APPROVED CHAPTER MODULES:

OFFICIAL PUBLICATION METADATA & VERIFICATION

Book Code: MOE-FORMULA-COMPENDIUM
Pages: 88 Official Pages | Size: 9.4 MB
Publisher: Ministry of Education Examination Authority - Egypt
Legal Deposit: 2025/14892 - ISBN: 978-977-12-7305-79
Verification URL: https://moe.gov.eg/ar/elearning-content/

MOE-FORMULA-

MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The Official Ministry Concepts & Formula Compendium ( ) is the authorized reference document provided to students during the Thanaweya Amma and EG-Baccalaureate national examinations. It contains every statutory mathematical formula, identity, standard integral, differential equation model, and equilibrium law permitted during state examinations.

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30%	**40%**	**30%**
Knowledge & Recall	**Understanding & Application**	**Higher Order Thinking (HOTS)**
Definitions, laws & formulas	Calculations, proofs & models	Analysis, synthesis & non-routine

3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal: https://moe.gov.eg/ar/elearning-content/. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: Authorized for All Senior Secondary Examination Sessions | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Pure Mathematics: Algebra & Combinatorics Section	0 wks	15 pts	25.0%	Counting, Binomial, Complex, Matrices
Pure Mathematics: Solid Analytical Geometry Section	0 wks	15 pts	25.0%	3D Lines, Planes, Vectors, Spheres
Pure Mathematics: Differential & Integral Calculus Section	0 wks	15 pts	25.0%	Trig/Log/Exp Derivatives, Standard Integrals, Areas, Volumes
Applied Mathematics: Statics & Dynamics Section	0 wks	15 pts	25.0%	Friction, Moments, Equilibrium, Newton Laws, Work & Energy

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism)

Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.

Engineering Sciences (Civil, Mechanical, Electrical)

Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.

Computer Science & Algorithmics

Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: SECTION 1: PURE MATHEMATICS FORMULA REGISTRY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Immediate recall and error-free substitution into all ministerial standard formulas.
- [+] Verify algebraic constraints before applying specialized formulas.

B. Statutory Theorems & Analytical Principles

All formulas in this compendium are verified by the National Center for Examinations and Educational Evaluation (NCEEE).

C. Standard Mathematical Expressions & Operational Formulas

- > $nPr = n! / (n - r)!$
- > $nCr = n! / [r!\,(n - r)!]$
- > $nCr / nC(r-1) = (n - r + 1) / r$
- > $T_{(r+1)} = nCr\, a^{(n-r)}\, b^r$
- > Euler Identity: $e^{(i \theta)} = \cos \theta + i \sin \theta$
- > De Moivre: $(\cos \theta + i \sin \theta)^n = \cos(n \theta) + i \sin(n \theta)$
- > Omega: $\omega^3 = 1$, $1 + \omega + \omega^2 = 0$, $\omega - \omega^2 = \pm i \sqrt{3}$
- > $A^{(-1)} = (1 / |A|)\, \mathrm{Adj}(A)$

D. National Examination Pitfalls & Common Student Errors

[!] Writing formulas from memory rather than verifying exact indices and signs against this official sheet.

CHAPTER SPECIFICATION: SECTION 2: APPLIED MATHEMATICS FORMULA REGISTRY

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

[+] Direct lookup of mechanics equilibrium and dynamics principles during timed national examinations.

B. Statutory Theorems & Analytical Principles

Statics and dynamics equations verified under SI and gravitational metric standards.

C. Standard Mathematical Expressions & Operational Formulas

```
> F_s = mu_s * R (limiting friction)
> R' = R sec lambda, tan lambda = mu_s
> M_O = r x F, |M_O| = F * L
> Parallel Forces: F1 * x1 = F2 * x2
> Center of Mass: X_G = Sum(m_i x_i) / Sum(m_i)
> v = dx/dt, a = dv/dt = v (dv/dx)
> F = m a, W = Integral F dx
> T = 1/2 m v^2, Delta T = W
> P = F . v, 1 HP = 735 W = 75 kg.wt.m/s
```

D. National Examination Pitfalls & Common Student Errors

[!] Using non-matching units when combining statics moments and dynamics forces.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

PURE MATH FAST REFERENCE
`cos^2 x + sin^2 x = 1`
`1 + tan^2 x = sec^2 x`
`1 + cot^2 x = csc^2 x`
`sin(2x) = 2 sin x cos x`
`cos(2x) = cos^2 x - sin^2 x = 2 cos^2 x - 1 = 1 - 2 sin^2 x`
`d/dx [tan x] = sec^2 x, d/dx [sec x] = sec x tan x`
`Integral (f'(x) / f(x)) dx = ln|f(x)| + C`
`Integral e^(k x) dx = (1/k) e^(k x) + C`

MECHANICS FAST REFERENCE
`F_max = mu_s R`
`M_O = r x F`
`Equilibrium: Sum Fx = 0, Sum Fy = 0, Sum M = 0`
`Elevator: N = m(g +- a)`
`Atwood: a = (m1 - m2)g / (m1 + m2)`
`Impulse I = F Delta t = m Delta v`
`Work-Energy: 1/2 m v^2 - 1/2 m v0^2 = W`

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	Authorized for All Senior Secondary Examination Sessions
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- Keep this compendium accessible on your exam desk at all times.
- No handwritten annotations or additional sheets may be attached to this official booklet.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

`Official Document Code: MOE-FORMULA-COMPENDIUM`
Curriculum Cycle: 2025/2026 Academic Year | National Edition
`Official Portal: https://moe.gov.eg/ar/elearning-content/`